

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
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S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection
- 2.



Project & Bridge Details

State – Rajasthan
Bridge No. – 903/2
Bridge Chainage – 1016+930

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

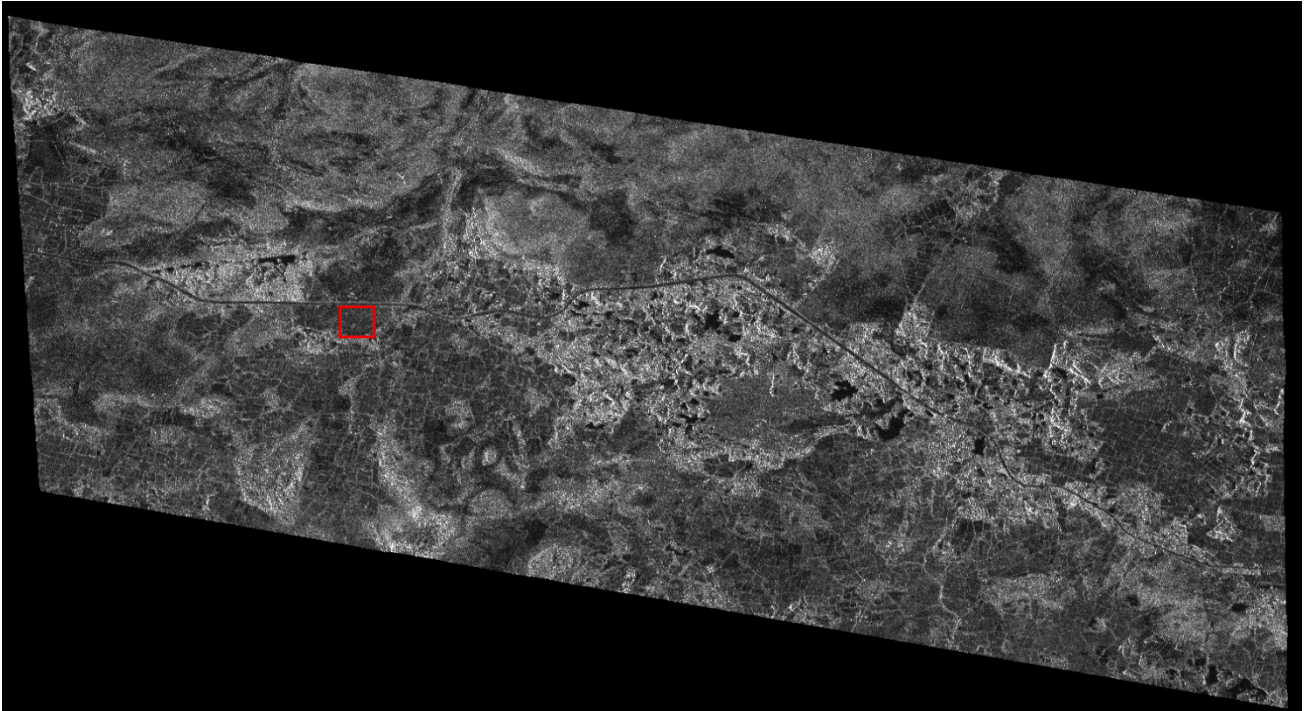
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

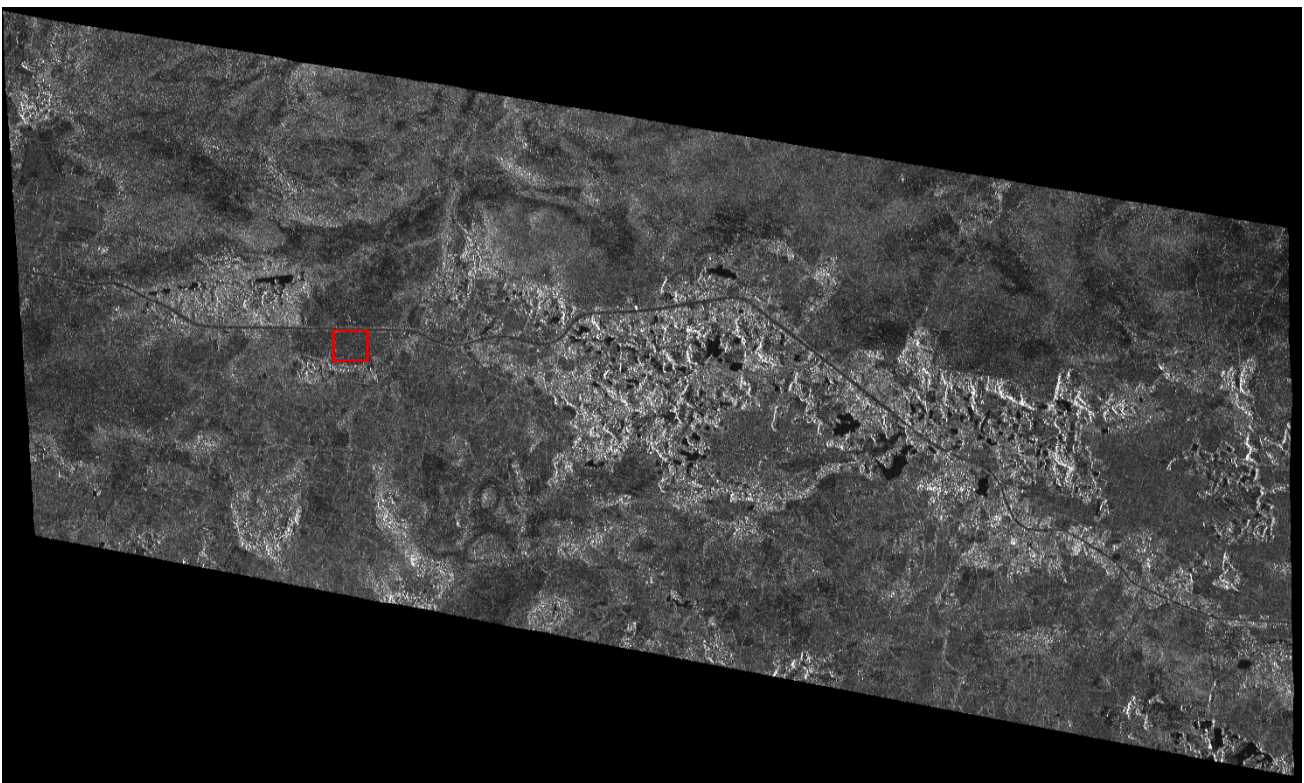
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

3. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



4. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-1016+930

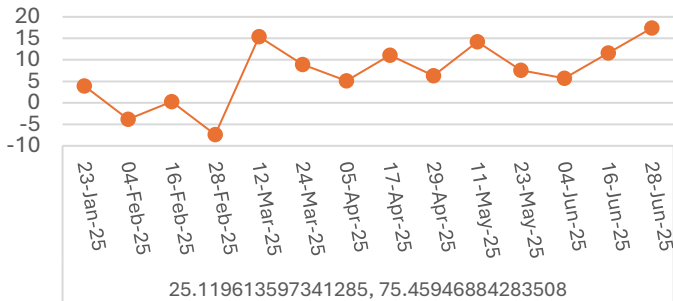
Length – 33.57M



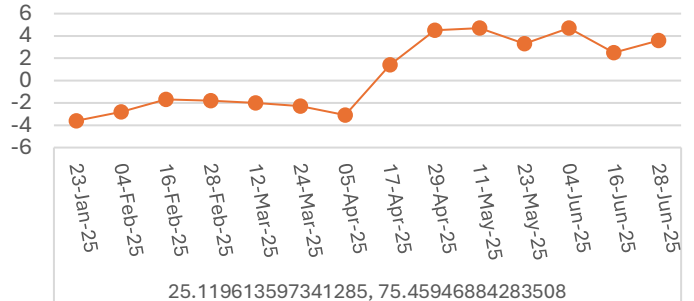
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
P2		Pier 2	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

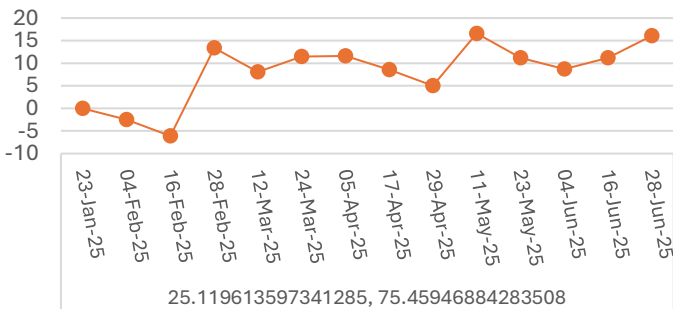
Displacement @ SR A1 LHS VV



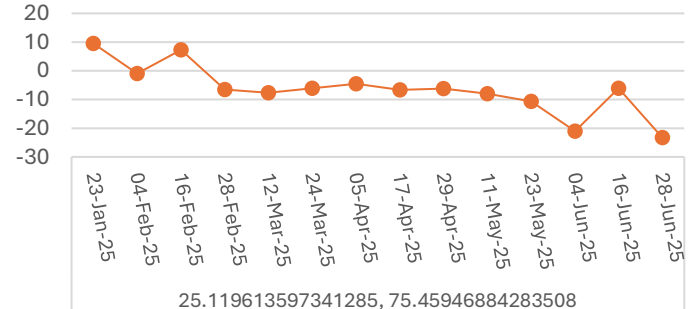
Displacement @ MR A1 LHS VV



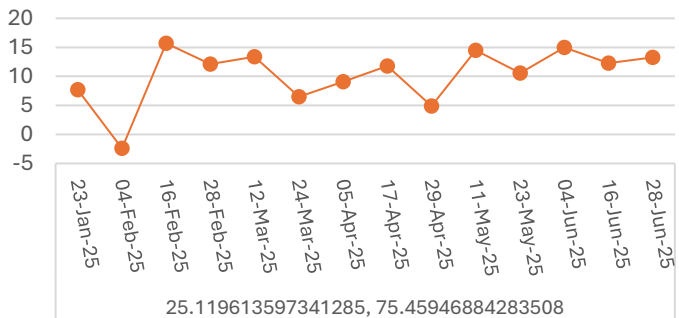
Displacement @ MR A1 RHS VV



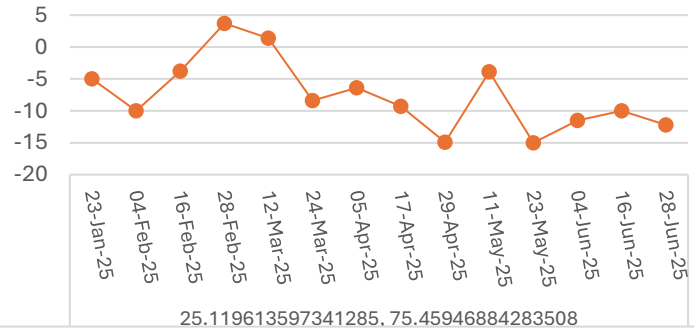
Displacement @ SR A1 RHS VV



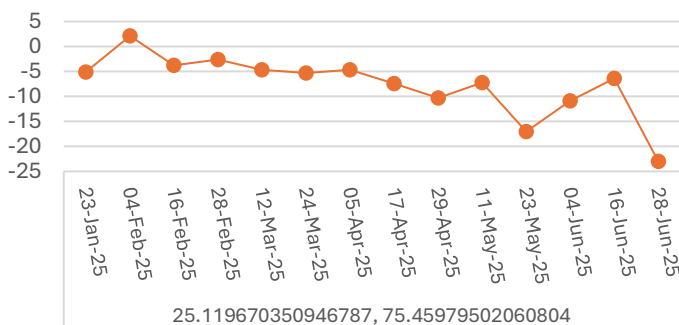
Displacement @ SR A2 LHS VV



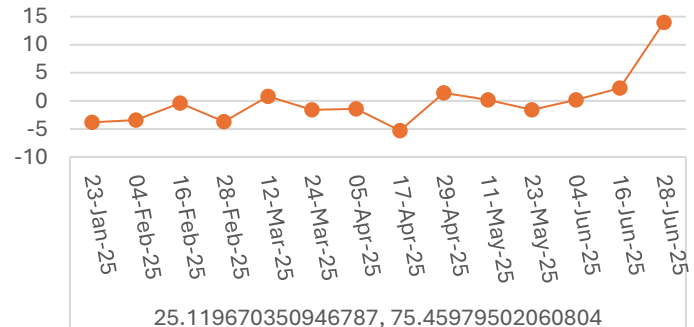
Displacement @ MR A2 LHS VV



Displacement @ MR A2 RHS VV

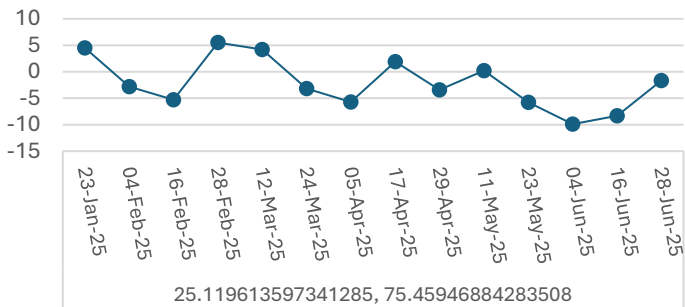


Displacement @ SR A2 RHS VV

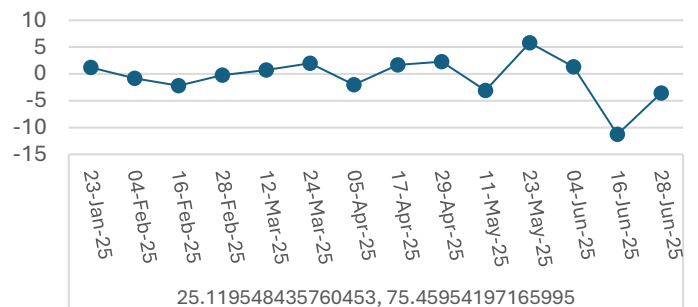


3.2 VH Displacement from 23 Jan 2025 To 28 June 2025

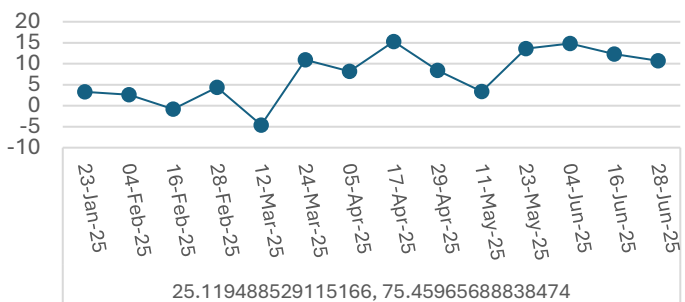
Displacement @ SR A1 LHS VH



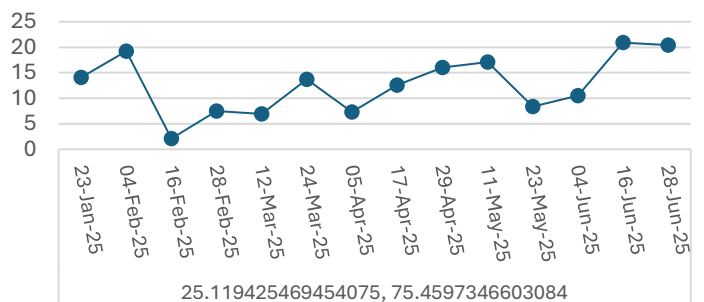
Displacement @ MR A1 LHS VH



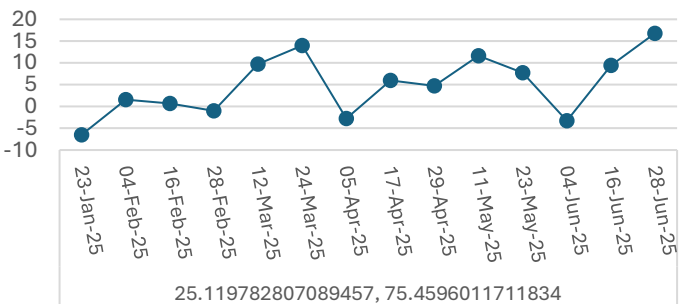
Displacement @ MR A1 RHS VH



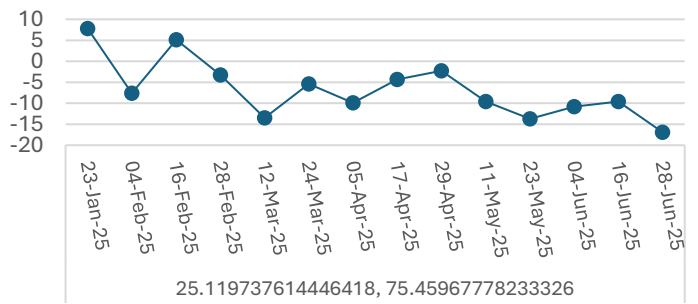
Displacement @ SR A1 RHS VH



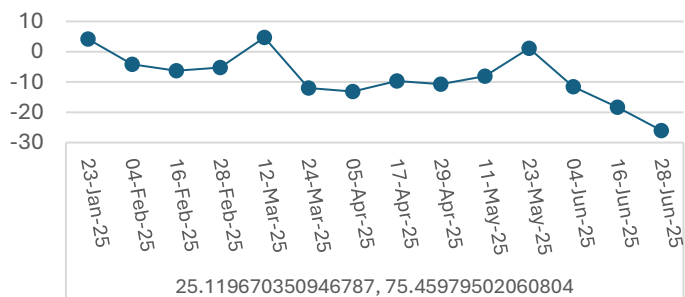
Displacement @ SR A2 LHS VH



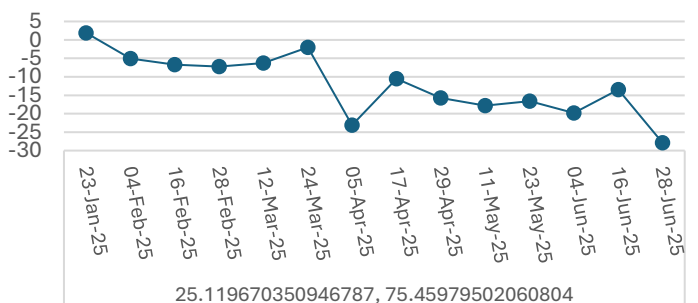
Displacement @ MR A2 LHS VH



Displacement @ MR A2 RHS VH



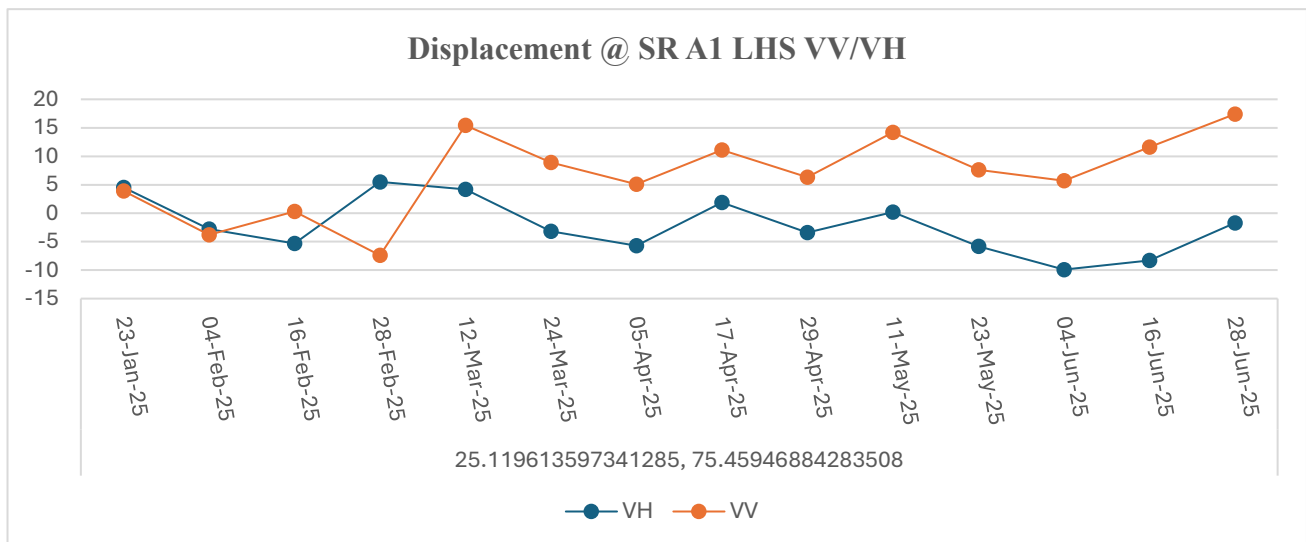
Displacement @ SR A2 RHS VH



3.3 SAR Based Analysis at Service Road Abutment 1 LHS:



SR LHS A1 (25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is predominantly positive throughout the monitoring period (Jan–Jun 2025), showing moderate to high variability.
- **Horizontal displacement (VH)** Horizontal displacement shows greater fluctuation and frequent negative values, ranging approximately from –12 mm to +5 mm.

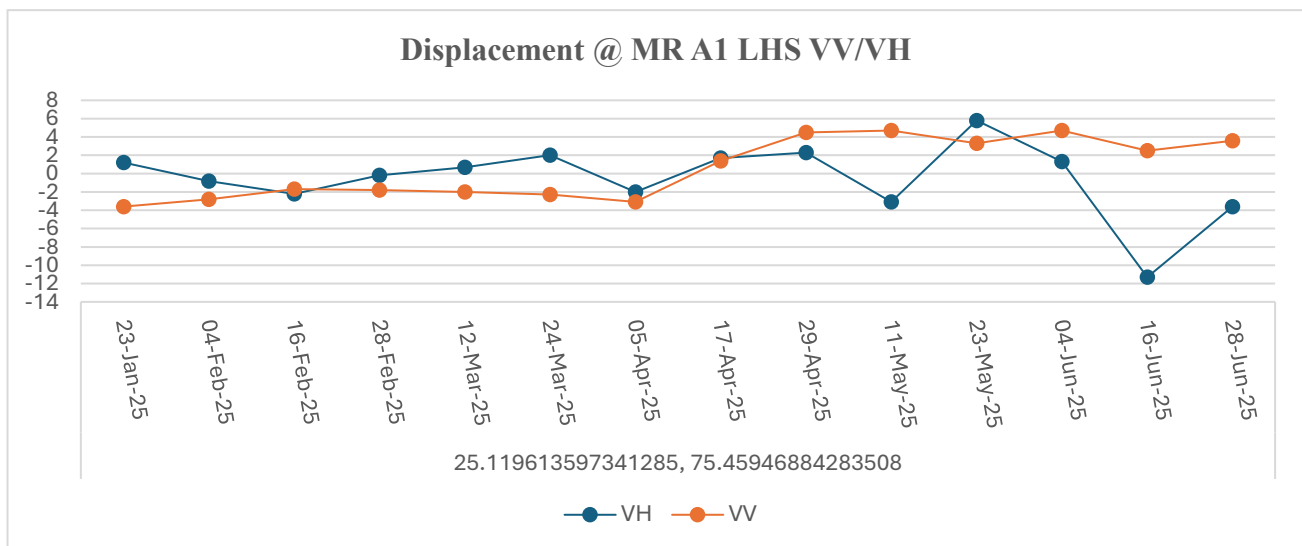
Critical Events & Anomalies:

- **January–February 2025:** VH transitions from slightly positive to negative values (down to ~–5 mm), while VV remains mildly positive.
- **April 2025:** Both VH and VV increase together (VH approaching 0 to +2 mm, VV around +10–12 mm), indicating a combined lateral–vertical displacement phase, possibly linked to external forcing (e.g., loading or hydrological effects).
- **June 2025:** VV reaches one of its highest values (+18 mm) while VH partially recovers but remains negative (–3 mm).

3.4 SAR Based Analysis at Main Road Abutment 1 LHS:



MR LHS A1 (25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits higher variability and frequent sign changes, ranging approximately from -12 mm to +6 mm.
- **Vertical displacement (VV)** Vertical displacement remains mostly positive from mid-February onward, with values generally ranging between -3 mm and +5 mm.

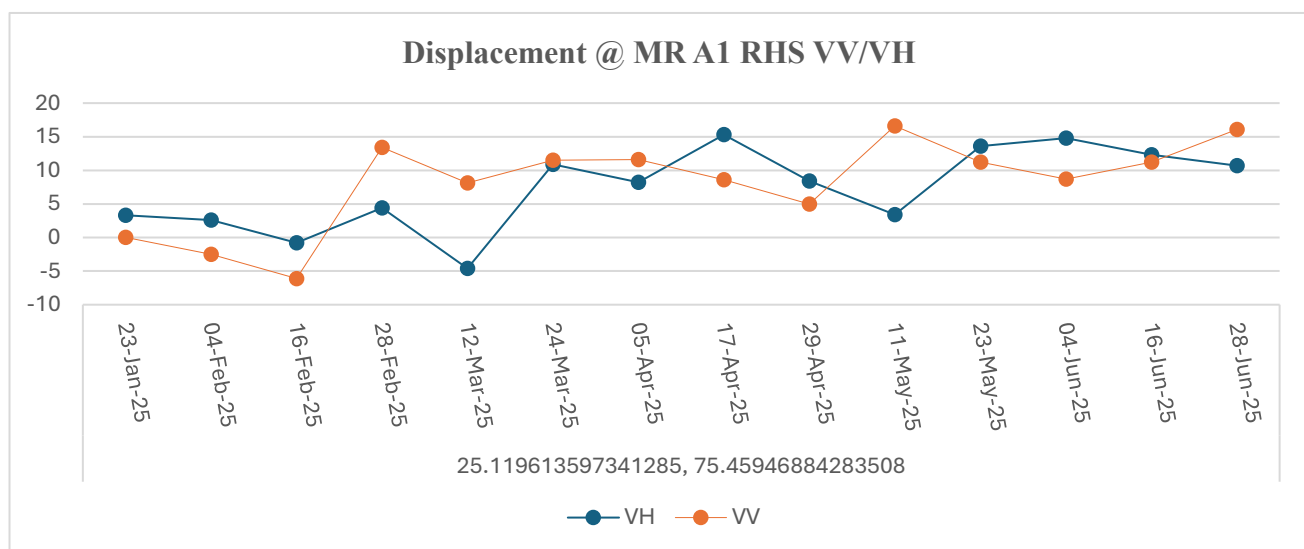
Critical Events & Anomalies:

- **February 2025:** Both VH and VV converge near zero displacement, indicating a short stabilization phase with minimal deformation in both directions.
- **April 2025:** VH drops back toward zero, and VV begins transitioning into positive values. This marks the onset of gradual vertical displacement dominance.
- **June 2025:** VH partially recovers (to -4 mm), and VV remains stable (+4 mm), suggesting lateral rebound with continued vertical consistency.

3.5 SAR Based Analysis at Main Road Abutment 1 RHS:



MR RHS A1 (25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a predominantly positive and strengthening trend from February through June 2025.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits moderate to high variability, ranging roughly between -2 mm and +15 mm.

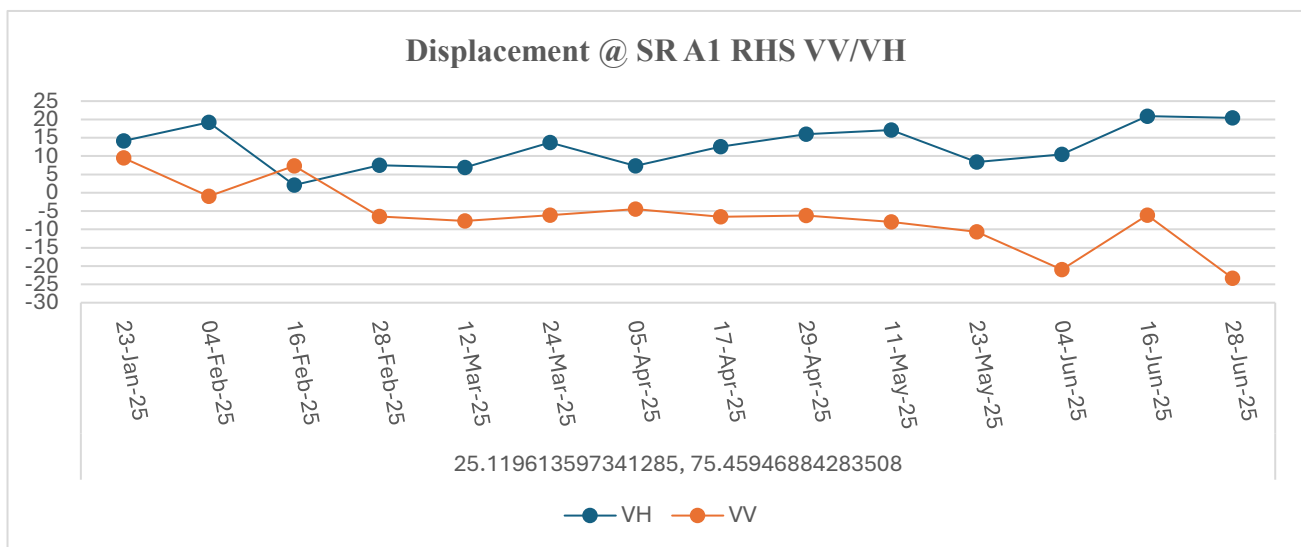
Critical Events & Anomalies:

- **January-February 2025:** VH remains slightly positive (+2–4 mm), while VV decreases into negative values (down to -4 mm). This indicates minor lateral movement concurrent with short-term vertical settlement.
- **April 2025:** VH reaches a local peak (+15 mm) while VV slightly decreases (+8 mm), implying lateral displacement dominance during this interval.
- **June 2025:** VV again approaches +16–17 mm, while VH remains elevated (+10–11 mm), indicating continued vertical accumulation with stable lateral movement.

3.6 SAR Based Analysis at Service Road Abutment 1 RHS:



SR RHS A1 (25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period (Jan–Jun 2025), indicating sustained downward movement.
- **Horizontal Displacement (VH)** Horizontal displacement remains consistently positive, ranging approximately from +5 mm to +20 mm, indicating continuous lateral movement in a dominant direction.

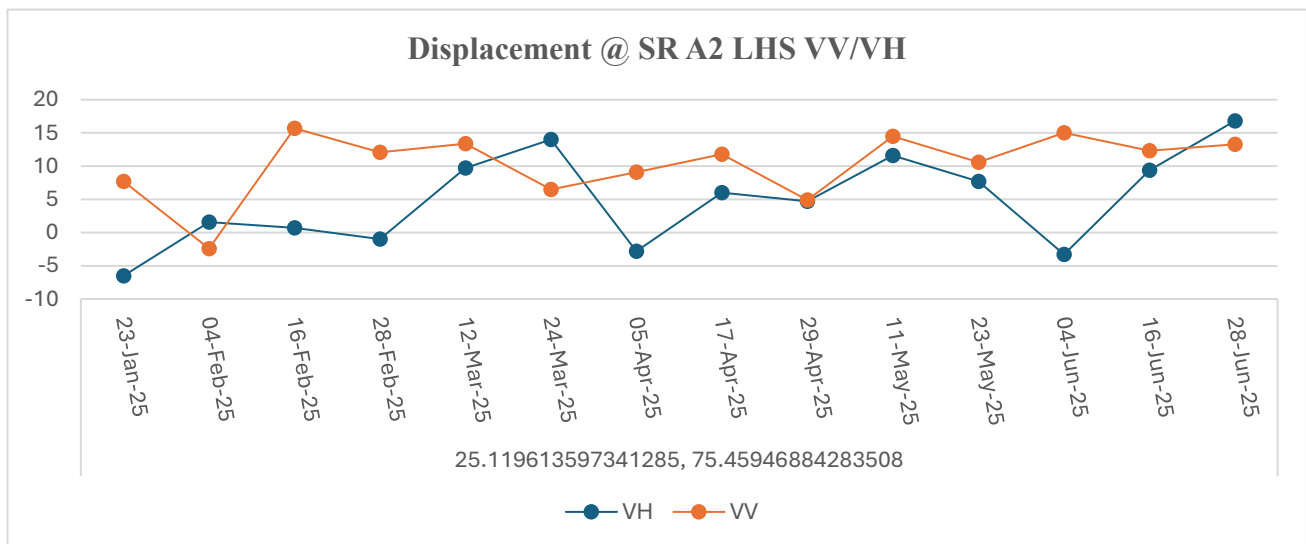
Critical Events & Anomalies:

- **February 2025:** VV shows a noticeable downward shift (–10 to –12 mm), while VH remains moderately positive (+8 to +10 mm), indicating increasing vertical settlement without corresponding lateral acceleration.
- **May 2025:** VV deepens further (–15 to –18 mm), while VH slightly dips before recovering, indicating short-term lateral relaxation under continued vertical settlement.
- **June 2025:** VV reaches its maximum downward displacement (–22 to –25 mm), while VH peaks around +20 mm.

3.7 SAR Based Analysis at Service Road Abutment 2 LHS:



SR LHS A2(25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement exhibits bidirectional variability, ranging approximately from –6 mm to +16 mm. While generally positive, intermittent negative excursions indicate short-lived reversals in lateral movement, suggesting sensitivity to transient loading or environmental effects.
- **Vertical Displacement (VV)** Vertical displacement remains consistently positive throughout the monitoring period (Jan–Jun 2025), indicating sustained uplift behaviour.

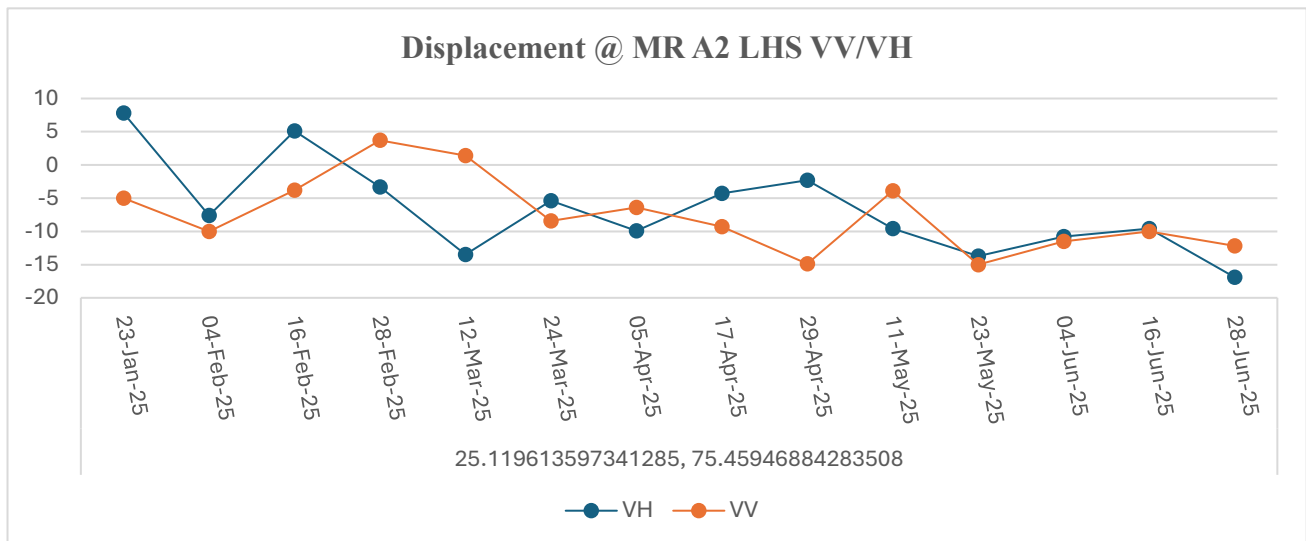
Critical Events & Anomalies:

- **February 2025:** VV rises sharply to around +16 mm, while VH oscillates near neutral (–1 to +2 mm), reflecting strong vertical response with minimal lateral contribution.
- **June 2025:** VH peaks near +17 mm, while VV stabilizes around +13–15 mm, indicating sustained lateral movement under continued vertical uplift.

3.8 SAR Based Analysis Main Road Abutment 2 LHS:



MR LHS A2(25.119613597341285, 75.45946884283508)



Key Observations:

Displacement Magnitude Patterns:

- Vertical Displacement (VV)** Vertical displacement remains predominantly negative throughout the monitoring period (Jan–Jun 2025), indicating sustained downward movement.
- Horizontal Displacement (VH)** Horizontal displacement shows high variability, fluctuating between approximately +8 mm and –17 mm.

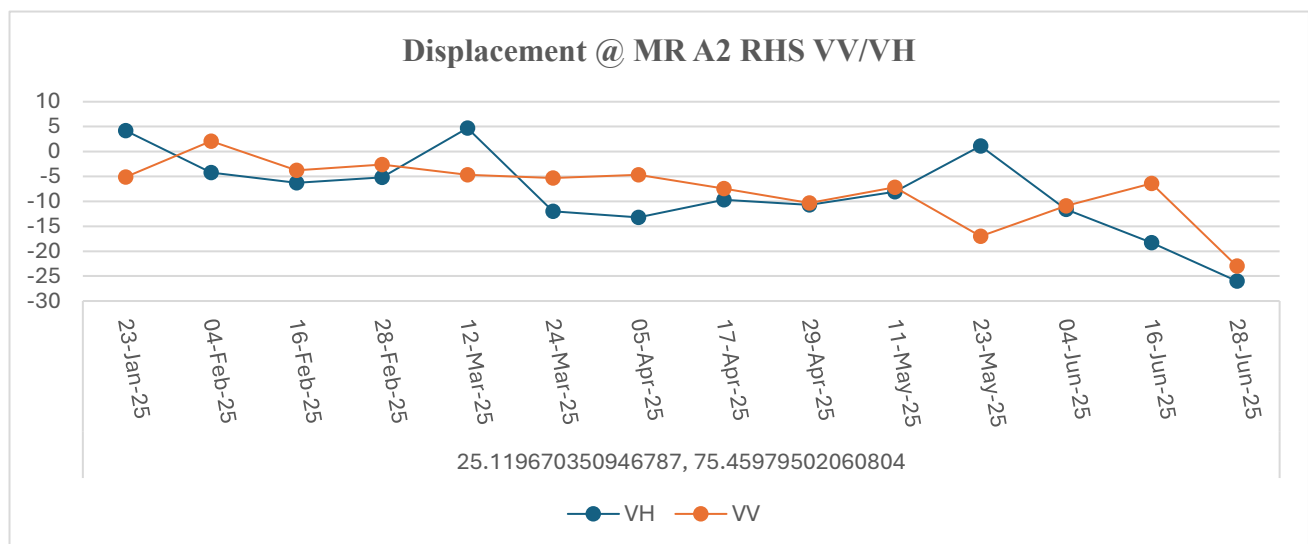
Critical Events & Anomalies:

- February 2025:** Both VH and VV fluctuate sharply. VV briefly rises toward –2 mm before dropping again, while VH oscillates between positive and negative values, reflecting unstable transitional behaviour.
- June 2025:** VH reaches its most negative level (–17 mm), while VV stabilizes near –10 to –12 mm, reflecting continued lateral progression under stabilized vertical settlement.

3.9 SAR Based Analysis at Main Road Abutment 2 RHS:



MR RHS A2 (25.119670350946787, 75.45979502060804)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period (Jan–Jun 2025), indicating continuous downward movement.
- **Horizontal Displacement (VH)** Horizontal displacement shows significant variability, ranging approximately from +5 mm to –25 mm. The overall trend transitions from mild positive movement early in the monitoring period to pronounced negative displacement toward the end, indicating a directional shift in lateral response.

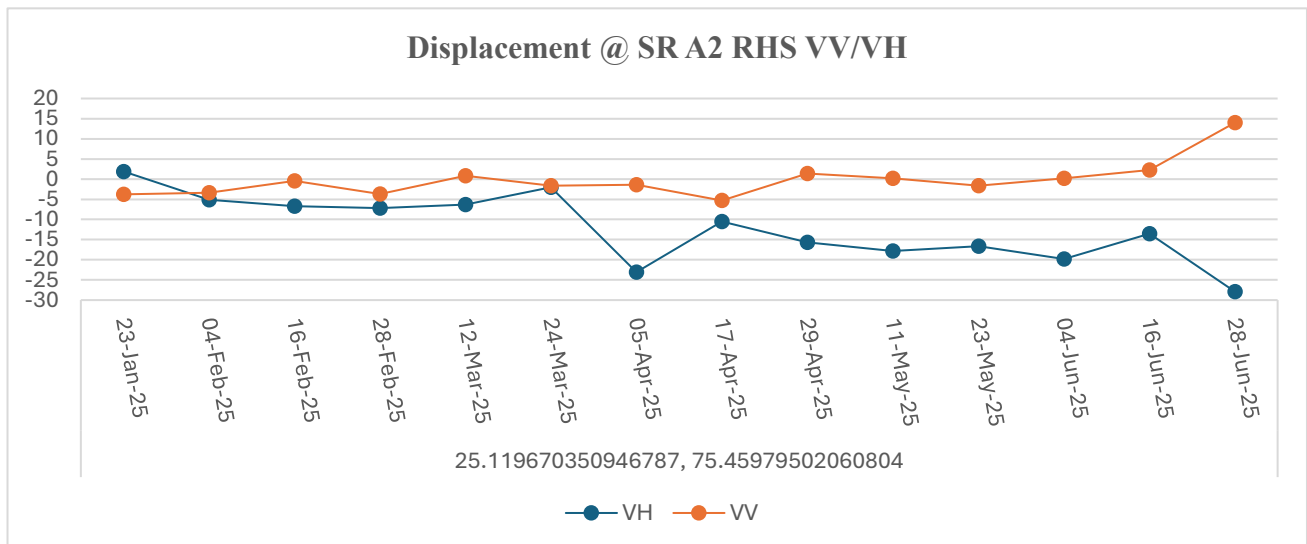
Critical Events & Anomalies:

- **February 2025:** VH fluctuates around neutral (–2 to +4 mm), while VV remains moderately negative (–5 to –8 mm), reflecting transitional behaviour with limited directional dominance.
- **June 2025:** VH reaches its lowest value (–25 mm), while VV also declines further (–22 mm), marking the most critical deformation phase observed in the monitoring period.

3.10 SAR Based Analysis at Service Road Abutment 2 RHS:



SR RHS A2 (25.119670350946787, 75.45979502060804)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a predominantly positive trend throughout the monitoring period (Jan–Jun 2025), indicating sustained uplift.
- **Horizontal Displacement (VH)** Horizontal displacement remains consistently negative, ranging approximately from –5 mm to –28 mm. The magnitude increases steadily over time, indicating continuous lateral movement in a dominant direction with limited recovery.

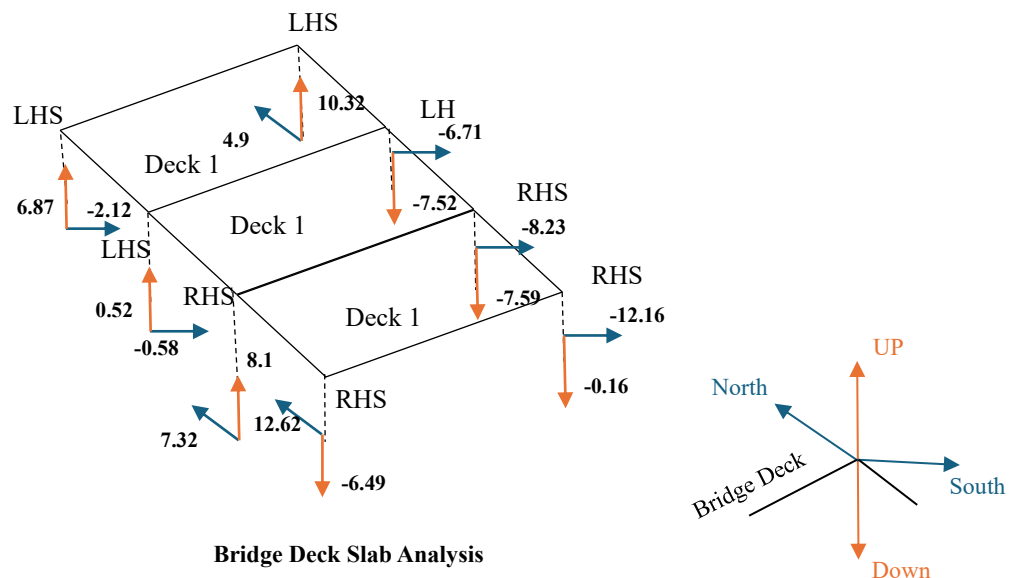
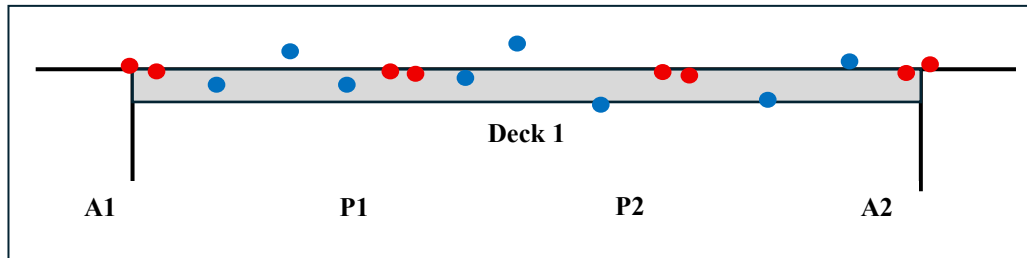
Critical Events & Anomalies:

- **Feb 2025:** VH gradually deepens to around –10 mm, while VV fluctuates mildly between –2 mm and +4 mm, suggesting early-stage lateral progression under relatively stable vertical conditions.
- **May 2025:** VH continues trending negatively (–15 to –18 mm), while VV increases steadily toward +8 mm, indicating renewed vertical uplift alongside sustained lateral movement.
- **June 2025:** VH reaches its most negative value (–28 mm), while VV peaks near +14 mm, marking a phase of simultaneous high lateral displacement and vertical uplift

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement across both LHS and RHS regions, indicating differential structural response rather than uniform movement. The maximum upward displacement is observed at the upper LHS deck region, reaching approximately +10.3 mm, indicating localized uplift

Horizontal Displacement Behaviour:

- Horizontal displacement exhibits clear directional variation across the deck. The LHS region shows moderate lateral movement, with values ranging between -2.1 mm and +6.9 mm, indicating relatively stable lateral behaviour. The RHS region, however, experiences more pronounced horizontal displacement, with values reaching approximately -12.2 mm at the lower deck level and -8.2 mm at the upper deck region

CONCLUSION:

- The bridge deck slab demonstrates a non-symmetric deformation pattern, characterized by moderate uplift on the LHS and pronounced lateral displacement on the RHS. Overall, the deformation pattern highlights the need for continued monitoring, particularly at RHS locations where combined vertical settlement and lateral movement are most pronounced.